

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

M. Tech. Mechanical (CAD/CAM) Semester-II University Examination, May-2011

ME-708 FINITE ELEMENT METHODS

Date: 17/05/2011, Tuesday

Time: 10:00 a.m. to 01:00 p.m.

Maximum Marks:70

Instructions:

1. Attempt **all** the questions.
2. Figures to the right indicate **full** marks.
3. Make suitable assumptions and draw neat figures wherever if required.

SECTION-I

Q1 Answer the following (Any Three) [15]

- (a) List the applications of finite element analysis in engineering field.
- (b) Describe the properties of global stiffness matrix.
- (c) Explain the Elimination approach with necessary derivation.
- (d) Explain the convergence requirement in FEM.

Q2 Prove following relationship for quadratic shape function and derive the equations for N_1 , N_2 and N_3 . [08]

$$\xi = \frac{2(x - x_3)}{x_2 - x_1}$$

OR

Q2 Determine the support reactions forces at two ends of the bar shown in Fig. [08]

1. Following data are given.

$$P = 8 \times 10^4 \text{ N,}$$

$$E = 2.0 \times 10^4 \text{ N / mm}^2,$$

$$A = 250 \text{ mm}^2,$$

$$L = 150 \text{ mm,}$$

$$\Delta = 1.4 \text{ mm}$$

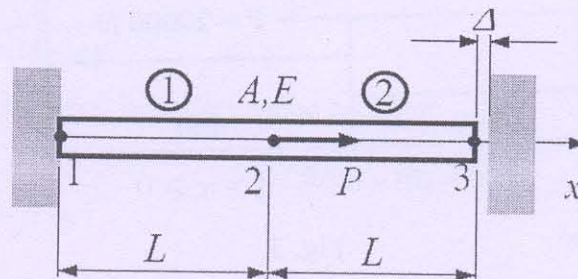


Fig.1

Q3 (a) Discuss the effect of thermal consideration in 1D spar elements. [04]

Q3 (b) Derive the following stiffness matrix in case of truss elements.

[08]

$$\mathbf{K} = \frac{A_e E_e}{L_e} \begin{bmatrix} l^2 & lm & -l^2 & -lm \\ lm & m^2 & -lm & -m^2 \\ -l^2 & -lm & l^2 & lm \\ -lm & m^2 & lm & m^2 \end{bmatrix}$$

OR

Q3 (a) Fig. 2 shows a cluster of five springs. One end of the assembly is fixed and a force of 800 N is applied at the end. Using the finite element method, determine:

[06]

- The deflection of each spring
- The reaction forces at support.

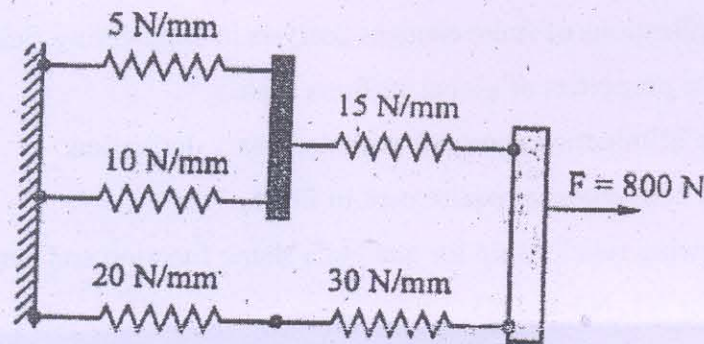


Fig. 2

Q3 (b) Consider the bar shown in Fig. 3. An axial load $F=20000\text{N}$ is as shown. Using the finite element method find the following:

[06]

- Determine the nodal displacement
- Determine the stresses in each section

$$A_1 = 2400 \text{ mm}^2 \quad A_2 = 600 \text{ mm}^2$$

$$E_1 = 70 \times 10^9 \text{ N/m}^2 \quad E_2 = 200 \times 10^9 \text{ N/m}^2$$

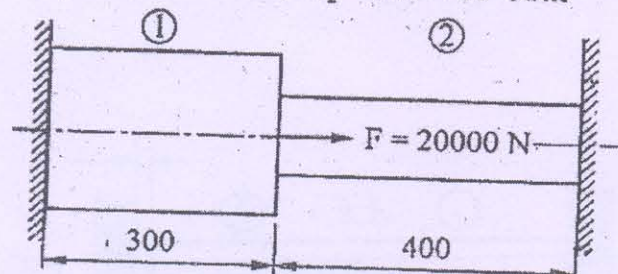


Fig. 3

SECTION-II

Q4 **Answer the following (Any Three)** **[15]**

- (a) Make an exhaustive list of finite element with its possible application.
- (b) Explain variational principles and finite element methods.
- (c) Explain plane stress and plane strain problems with suitable examples.
- (d) Explain multipoint boundary condition.

Q5 **Derive the expression for element mass matrices in dynamic problem.** **[08]**

OR

Q5 The stresses at node points 1, 2 and 3 are 90, 120 and 160 MPa respectively. **[08]**
Determine the stress at point P for the co ordinate as shown in **Fig. 4**.

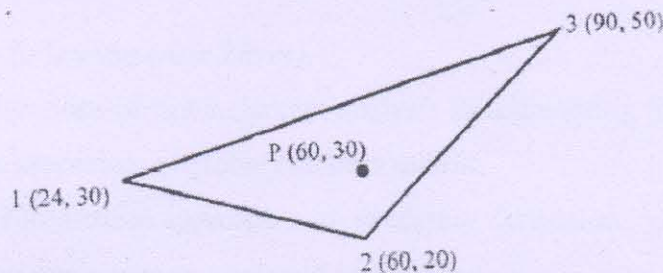


Fig.4

Q6 A Four bar truss is shown in **Fig. 5** Assuming that for each element, the **[12]**
cross-sectional area is 400 mm² and modulus of elasticity is 200 GPa.
Determine the deflections, reactions and stresses in each element.

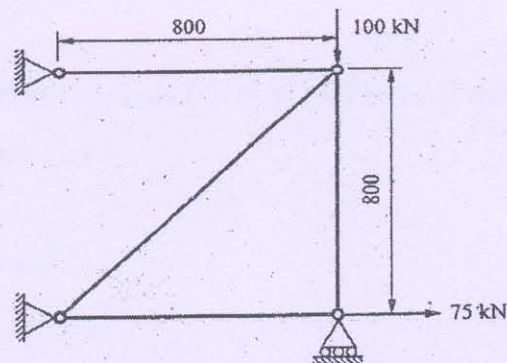


Fig. 5

OR

Q6 **Answer the following.**

Q6 (a) Consider the following boundary value problem: **[08]**

$$\frac{d^2 y}{dx^2} + u = 1 \quad 0 \leq x \leq 1 \quad \text{With } u(0) = 0 \text{ and } \frac{du}{dx} = 0 \text{ at } x = 1$$

Solve this equation using Rit'z method.

Q6 (b) Explain the nonlinear FEA and its application areas. **[04]**